

# OpenJLS

## JPEG-LS Lossless Encoder IP Core — Datasheet

### 1 Overview

OpenJLS is a JPEG-LS lossless encoder for FPGAs. It encodes single-component (grayscale) images at one pixel per clock with no external memory, streaming a standards-compliant JPEG-LS bitstream out over a standard ready/valid streaming handshake. Source code, license and the full verification suite are public at <https://github.com/isentropic-fpga/OpenJLS>. There's also a companion public repository with demo projects for OpenJLS at <https://github.com/isentropic-fpga/OpenJLS-Demos>.

**Table 1:** Key specifications.

|                    |                                                            |
|--------------------|------------------------------------------------------------|
| <b>Standard</b>    | JPEG-LS (ISO/IEC 14495-1, ITU-T T.87)                      |
| <b>Compression</b> | Lossless only                                              |
| <b>Components</b>  | Single (grayscale)                                         |
| <b>Pixel depth</b> | 8–16 bits (BITNESS)                                        |
| <b>Image size</b>  | Up to $65535 \times 65535$ px                              |
| <b>Throughput</b>  | 1 pixel / clock                                            |
| <b>Latency</b>     | 14 clock cycles                                            |
| $f_{max}$          | ~280 MHz at 12 bit (Xilinx xczu7eg-1)                      |
| <b>Interface</b>   | Ready/valid streaming (AXI4-Stream / Avalon-ST compatible) |
| <b>Memory</b>      | On-chip line buffer, one image row; no external RAM        |
| <b>Resources</b>   | ~6.4k LUT, ~2.0k FF + BRAM scaling with image width        |
| <b>Portability</b> | Vendor-agnostic VHDL                                       |

### 2 Revision and contact

**Table 2:** Revision history.

| Version | Date    | Notes                                                                     |
|---------|---------|---------------------------------------------------------------------------|
| 1.0     | 2026-06 | Initial release.                                                          |
| 1.1     | 2026-07 | Packaged IPs for Vivado.                                                  |
| 1.2     | 2026-07 | Isentropic branding; cores repackaged under the <b>isentropic</b> vendor. |
| 1.3     | 2026-09 | Frequency increased ~15%, LUT reduction ~16%, BRAM up 1 tile.             |

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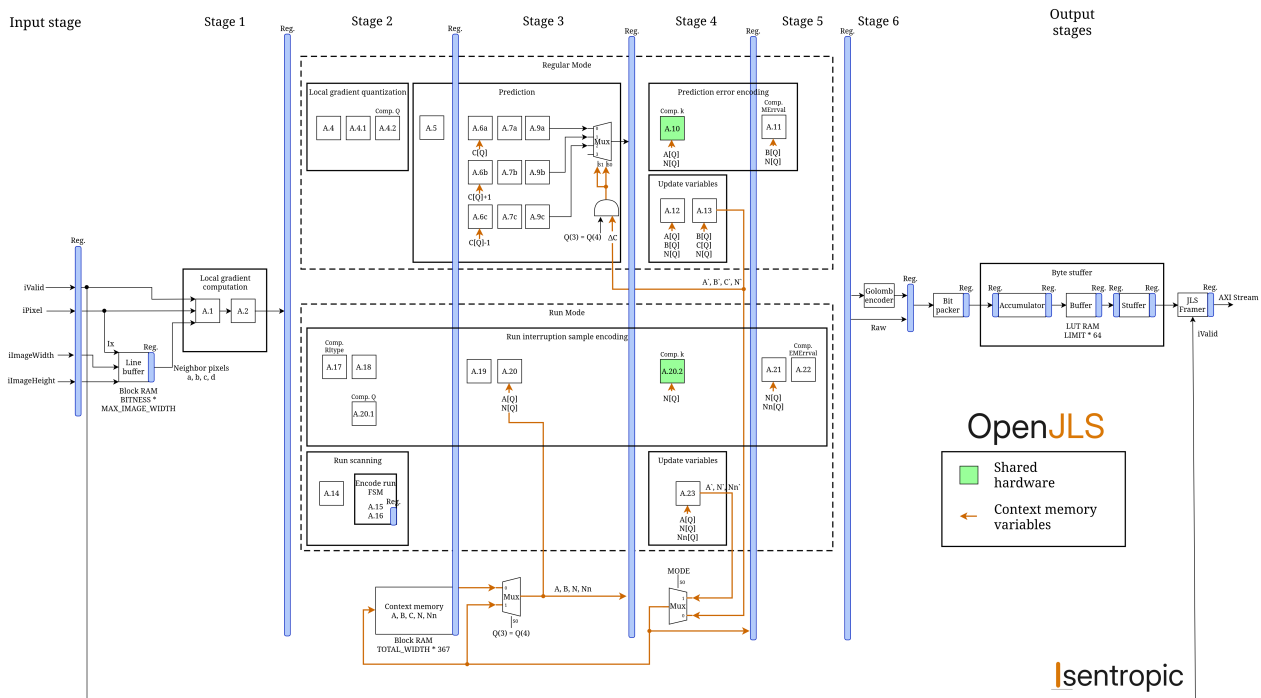
### 3 Functional description

The core is a fully pipelined encoder with a latency of 14 clock cycles: an input register, six processing stages (gradient computation, context modeling, MED prediction with bias correction, Golomb–Rice / run-length coding) and three internally registered output stages (bit packer, byte stuffer, framer).

End-of-image is derived internally from the dimensions provided at the ports `iImageWidth` and `iImageHeight`. The core never stalls — neither at EOI (end of image) nor at EOL (end of line) — which allows processing multiple images back-to-back. The only exception is when the downstream stalls long enough for the byte-stuffer buffer to fill, which in turn stalls the upstream.

While the pipeline latency is 14 cycles, the core does not emit an output beat every cycle: the output stages accumulate compressed bits and assert `oValid` only once a full `OUT_WIDTH` word is ready. As the bitstream is compressed, these output beats are infrequent relative to the one-pixel-per-clock input.

**Figure 1:** Encoder pipeline architecture.



A [full-resolution version](#) of this diagram is available in the repository.

## 4 Generics

The core is configured by the four generics in Table 3.

**Table 3:** Configuration generics.

| Generic          | Range   | Description                                      |
|------------------|---------|--------------------------------------------------|
| BITNESS          | 8–16    | Pixel bit depth.                                 |
| MAX_IMAGE_WIDTH  | 4–65535 | Largest width supported; sets line-buffer depth. |
| MAX_IMAGE_HEIGHT | 1–65535 | Largest height supported.                        |
| OUT_WIDTH        | 48–1024 | Output bus width in bits; multiple of 8.         |

MAX\_IMAGE\_WIDTH and MAX\_IMAGE\_HEIGHT fix the *compile-time* maximum image size: they size the on-chip line buffer and the dimension counters, so a larger maximum costs more BRAM. They do not select the size of a given image — the dimensions of each encoded image are chosen at *run time* through the `iImageWidth` and `iImageHeight` ports (Table 4), which accept any value from the minimum up to the configured maximum.

## 5 Ports

All ports are synchronous to the rising edge of `iClk` — the core is a single clock domain. The streaming ports use a plain ready/valid handshake; in Table 4 the **AXIS** column gives the one-to-one AXI4-Stream signal mapping for that ecosystem (Avalon-ST maps the same way at `readyLatency` = 0).

**Table 4:** Port list and AXI4-Stream mapping.

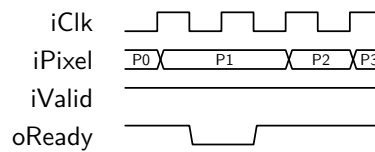
| Port         | Dir | Width       | AXIS   | Role                                                |
|--------------|-----|-------------|--------|-----------------------------------------------------|
| iClk         | in  | 1           | —      | Clock.                                              |
| iRst         | in  | 1           | —      | Synchronous reset, active high; latches dimensions. |
| iImageWidth  | in  | 16          | —      | Image width, pixels (configuration).                |
| iImageHeight | in  | 16          | —      | Image height, pixels (configuration).               |
| iValid       | in  | 1           | TVALID | Pixel valid.                                        |
| iPixel       | in  | BITNESS     | TDATA  | Pixel data.                                         |
| oReady       | out | 1           | TREADY | Input ready.                                        |
| oData        | out | OUT_WIDTH   | TDATA  | Bitstream beat.                                     |
| oValid       | out | 1           | TVALID | Output valid.                                       |
| oKeep        | out | OUT_WIDTH/8 | TKEEP  | Valid-byte mask on final beat.                      |
| oLast        | out | 1           | TLAST  | End of image.                                       |
| iReady       | in  | 1           | TREADY | Downstream ready.                                   |

**IMPORTANT:**

- `iImageWidth` and `iImageHeight` are only sampled while `iRst` is asserted ('1').
- If an invalid value is provided to `iImageWidth`, that is `iImageWidth` > `MAX_IMAGE_WIDTH` or `iImageWidth` < 4, the encoder uses `MAX_IMAGE_WIDTH` as the image width.
- If an invalid value is provided to `iImageHeight`, that is `iImageHeight` > `MAX_IMAGE_HEIGHT` or `iImageHeight` < 1, the encoder uses `MAX_IMAGE_HEIGHT` as the image height.
- If you don't intend to change the image dimensions at run-time, wire both inputs to 0 so that the maximum values are used.
- Both ports are a fixed 16 bits regardless of the `MAX_IMAGE_*` generics — every representable value is checked, so any out-of-range value is caught and replaced by the maximum.

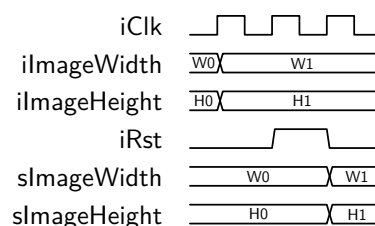
## 6 Timing and protocol

**Pixel input.** Pixels are streamed in scan order — the first row left to right, then the second row, and so on — each an unsigned value on `iPixel`. A pixel transfers on every cycle where `iValid` and `oReady` are both high. The core asserts `oReady` continuously and deasserts it only when the FIFO in byte stuffer fills under downstream backpressure. Figure 2 shows a transfer with a brief downstream stall.

**Figure 2:** Pixel-input handshake.

In Figure 2, P2 is held flat across the stall and accepted when `oReady` returns high; no pixel is dropped while `oReady` is low.

**Reset and configuration.** `iRst` is level-sensitive (hold it high for at least one clock) and clears the pipeline. The image dimensions are captured only while `iRst` is high: present `iImageWidth` and `iImageHeight`, hold them stable across the reset pulse, and they latch on the rising edge (Figure 3). A reset is therefore needed only to capture the dimensions — before the first image and whenever the image size changes; with the size unchanged, images encode back-to-back.

**Figure 3:** Reset and dimension capture.

In Figure 3, the new size (W1/H1) is driven for several cycles, then `iRst` pulses high; the dimensions register on the rising edge while `iRst` is high.

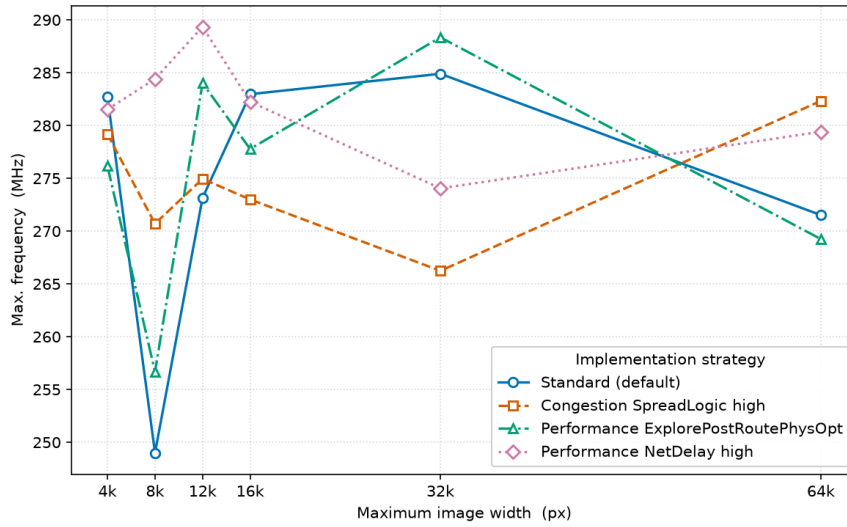
**Output.** Beats appear on `oData/oValid` subject to `iReady`; `oLast` marks the final beat (EOI) and `oKeep` flags its valid bytes. The bitstream is byte-serial and MSB-first: the earliest byte occupies the most-significant byte of `oData`. Pipeline latency is 14 clock cycles.

## 7 Resources and performance

Characterized on Xilinx Zynq UltraScale+ `xczu7eg-fbvb900-1-e`, Vivado 2025.2, using the generics `BITNESS = 12` (unless stated otherwise) and `OUT_WIDTH = 64`, RTL-only with no floorplanning.  $f_{max}$  is the true maximum, found by over-constraining the clock until timing fails. At one pixel per clock,  $\sim 280$  MHz is  $\sim 280$  Mpixel/s.

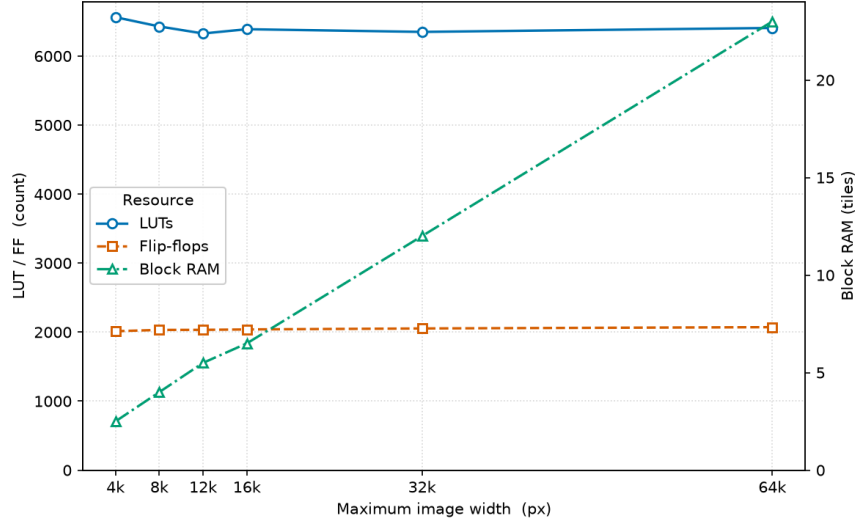
Figure 4 and Table 5 report  $f_{max}$  across `MAX_IMAGE_WIDTH`; Figure 5 and Table 6 report the resource usage; Figure 6 and Table 7 report both across `BITNESS`.

**Figure 4:** Maximum frequency vs maximum image width.



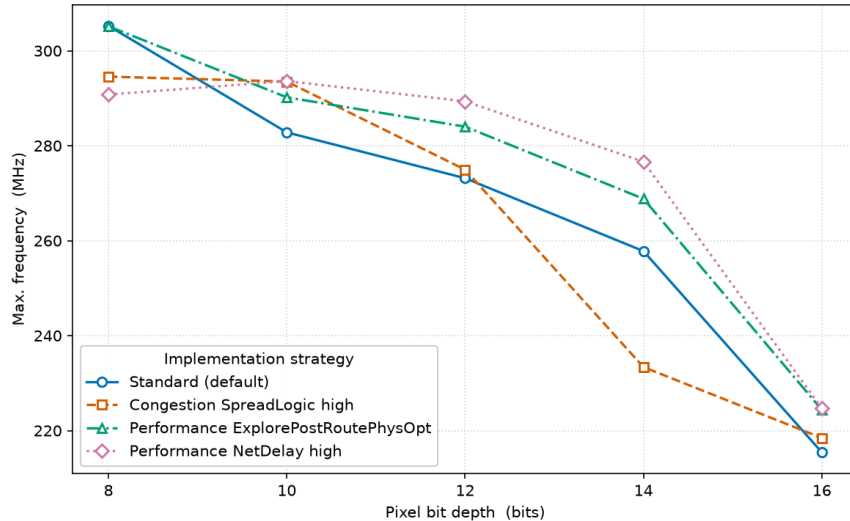
**Table 5:** Maximum frequency by maximum image width and strategy.

| MAX_IMAGE_WIDTH | Default      | EPRPO        | NetDelay     | Congestion   |
|-----------------|--------------|--------------|--------------|--------------|
| 4096            | <b>282.7</b> | 276.2        | 281.5        | 279.2        |
| 8192            | 249.0        | 256.7        | <b>284.4</b> | 270.7        |
| 12288           | 273.1        | 284.0        | <b>289.4</b> | 274.9        |
| 16384           | <b>283.0</b> | 277.8        | 282.2        | 273.0        |
| 32768           | 284.9        | <b>288.4</b> | 274.1        | 266.2        |
| 65535           | 271.5        | 269.2        | 279.4        | <b>282.3</b> |

**Figure 5:** Resource usage vs maximum image width.**Table 6:** Resource usage by maximum image width.

| MAX_IMAGE_WIDTH | LUT  | FF   | BRAM |
|-----------------|------|------|------|
| 4096            | 6563 | 2013 | 2.5  |
| 8192            | 6429 | 2032 | 4.0  |
| 12288           | 6327 | 2033 | 5.5  |
| 16384           | 6389 | 2039 | 6.5  |
| 32768           | 6350 | 2053 | 12.0 |
| 65535           | 6407 | 2072 | 23.0 |

In Table 5 frequencies are in MHz and the best strategy per row is in bold; the strategies are **Performance\_ExplorePostRoutePhysOpt** (EPRPO), **Performance\_NetDelay\_high** (NetDelay) and **Congestion\_SpreadLogic\_high** (Congestion). The design is congestion-bound, so the winning strategy shifts with image size; a given netlist is deterministic, but the per-size winner is placement-sensitive and can shift when the netlist changes, so the best-of band ( $\sim 282$ – $289$  MHz) is the headline figure; the per-size spread across strategies is 7–22 MHz, so a default run lands close to it at most sizes. Table 6 is for the default strategy; utilization is near-identical across strategies. Logic is essentially constant with image size; only the line-buffer BRAM scales with width and pixel depth.

**Figure 6:** Maximum frequency vs pixel bit depth (MAX\_IMAGE\_WIDTH = 12288).

**Table 7:** Maximum frequency and resource usage by pixel bit depth (`MAX_IMAGE_WIDTH = 12288`).

| BITNESS | Default      | EPRPO | NetDelay     | Congestion | LUT  | FF   | BRAM |
|---------|--------------|-------|--------------|------------|------|------|------|
| 8       | <b>305.3</b> | 305.2 | 290.8        | 294.6      | 5319 | 1772 | 3.5  |
| 10      | 282.8        | 290.2 | <b>293.6</b> | 293.5      | 5908 | 1902 | 4.0  |
| 12      | 273.1        | 284.0 | <b>289.4</b> | 274.9      | 6327 | 2033 | 5.5  |
| 14      | 257.8        | 268.8 | <b>276.6</b> | 233.4      | 6845 | 2153 | 6.5  |
| 16      | 215.4        | 224.4 | <b>224.7</b> | 218.4      | 7601 | 2287 | 7.5  |

Best-of  $f_{max}$  falls from  $\sim 305$  MHz at 8 bits to  $\sim 277$  MHz at 14 bits and  $\sim 225$  MHz at 16 bits. The byte stuffer is off the critical path at every depth; the limit is the context RAM read into the prediction-error register, whose carry chains widen with pixel depth. Resources are for the default strategy.

## 8 Integration example

```
u_enc : entity work.openjls_top
generic map (
  BITNESS => 8,
  MAX_IMAGE_WIDTH => 4096,
  MAX_IMAGE_HEIGHT => 4096,
  OUT_WIDTH => 64 )
port map (
  iClk => clk,
  iRst => rst,
  iImageWidth => img_w, -- latched on reset
  iImageHeight => img_h, -- latched on reset
  iValid => s_valid, -- pixel stream (AXIS slave)
  iPixel => s_data,
  oReady => s_ready,
  oData => m_tdata, -- bitstream (AXIS master)
  oValid => m_tvalid,
  oKeep => m_tkeep,
  oLast => m_tlast,
  iReady => m_tready );
```

Being a single entity with standard ready/valid ports, `openjls_top` can equally be dropped onto a block diagram and wired there, instead of instantiating it in HDL. The dimension ports are fixed at 16 bits (rather than sized from the generics) because Vivado's block-design port-width evaluator only handles literal arithmetic in port expressions.

## Xilinx IP cores

For the Vivado flow the core ships the following pre-packaged IPs

| IP Catalog name                           | Interfaces                                       |
|-------------------------------------------|--------------------------------------------------|
| OpenJLS Encoder (Native)                  | Plain Ready/Valid                                |
| OpenJLS Encoder (AXI4-Stream)             | AXI-Stream for the data, native pins for control |
| OpenJLS Encoder (AXI4-Stream + AXI4-Lite) | AXI-Stream for the data, AXI-Lite for control    |

To use simply add `Sources/Xilinx/ip_repo/` to the project's IP repositories and the three cores appear in the IP Catalog, ready to drop onto a block design.

## AXI4-Lite register map

The AXI4-Lite variant replaces the native control pins with a register bank (32-bit registers, word-aligned offsets):

| Offset | Name    | Access | Contents                                                      |
|--------|---------|--------|---------------------------------------------------------------|
| 0x00   | ID      | RO     | ASCII "OJLS" (0x4F4A4C53)                                     |
| 0x04   | VERSION | RO     | 0x00MMmmpp (major/minor/patch)                                |
| 0x08   | CAPS    | RO     | [7:0] BITNESS, [15:8] output bytes per beat (OUT_WIDTH/8)     |
| 0x0C   | MAXDIM  | RO     | [15:0] MAX_IMAGE_WIDTH, [31:16] MAX_IMAGE_HEIGHT              |
| 0x10   | WIDTH   | RW     | [15:0] image width (clamped on write)                         |
| 0x14   | HEIGHT  | RW     | [15:0] image height (clamped on write)                        |
| 0x18   | CTRL    | WO     | [0] APPLY — self-clearing, pulses the core reset              |
| 0x1C   | STATUS  | RO     | [0] BUSY (reset pulse active), [1] pixel-stream TREADY mirror |

The core samples the dimensions only while its reset is high, so reconfiguration is: write WIDTH/HEIGHT, set CTRL.APPLY. APPLY pulses the core reset for one clock while the registers are held stable; the AXI-Lite endpoint itself stays on `aresetn` only, so the bus never disappears mid-reconfigure. While the pulse is active STATUS.BUSY reads 1 and the pixel stream's TREADY is held low, so a stream started too early stalls instead of losing pixels — WIDTH/HEIGHT/CTRL writes while BUSY are dropped. Back-to-back images of unchanged dimensions need no APPLY.

WIDTH/HEIGHT writes are merged per WSTRB and clamped to the core's rule (out-of-range values become the MAX generic), so a readback always returns the value the core will actually use. Writes to RO or unmapped offsets are acknowledged (OKAY) and dropped; unmapped reads return zero.

For driver code there is a C header in the repository, [ojls\\_regs.h](#).

## 9 Conformance and verification

Verified by simulation with NVC using a layered, self-checking suite run at both RTL and post-synthesis (gate-level) stages:

- **OSVVM** — Per-module tests against independent T.87-derived reference models, top-level control-plane stress (reset injection, backpressure, back-to-back images, dimension fallback), and AXI wrapper tests (byte-exact AXI4-Stream pass-through at 8- and 12-bit, AXI4-Lite register map, live reconfiguration, mid-image abort), all with requirements tracking.
- **Coverage** — OSVVM functional coverage with NVC structural code coverage (99%+ statements).
- **Golden model** — Byte-exact vs the independent CharLS C++ encoder over 287 real and synthetic images, plus ISO/IEC 14495-1 vectors.
- **Design contracts** — Embedded PSL assertions (ready/valid handshake protocol, internal handshakes) on every run.
- **Post-synthesis** — The stress test and a golden-model subset re-run on the synthesized netlist.
- **Hardware-in-the-loop** — The full 287-image corpus encoded end-to-end on a PYNQ-Z2 FPGA and byte-compared against CharLS on real silicon. Produced in the demos repository [OpenJLS-Demos](#) by the EncodeOverEthernet project.

Table 8 is the latest regression snapshot.

**Table 8:** Latest regression snapshot.

| Suite                   | Status | Test/Cov | Summary                                                        |
|-------------------------|--------|----------|----------------------------------------------------------------|
| NVC code coverage       | info   | 99.8%    | Per-DUT-file statement breakdown                               |
| OSVVM suite             | PASS   | 100%     | 60 tests, 2 877 433 affirmations (module + top + AXI wrappers) |
| Golden model            | PASS   | 100%     | 287/287 images byte-exact vs CharLS                            |
| Post-synth OSVVM        | PASS   | 100%     | Control-plane stress on the gate-level netlist                 |
| Post-synth golden model | PASS   | 100%     | 156/156 images byte-exact vs CharLS                            |
| Hardware-in-the-loop    | PASS   | 100%     | 287/287 images byte-exact vs CharLS on PYNQ-Z2 silicon         |

**References.** ITU-T T.87 / ISO/IEC 14495-1; Y.M.Mert, *Key Architectural Optimizations for Hardware Efficient JPEG-LS Encoder*, IEEE 2018; [CharLS](#); [open-logic](#).